Forces

Interact & deform

1 Vectors & resultants

Vectors have magnitude + direction; scalars have magnitude only. Force in N; weight **$W = mg$.**

Resultant combines forces. Zero resultant → constant velocity, possibly zero. Resolve perpendicular components with scale drawings.

2 Work & springs

$W = Fs$; $F = ke$; $E_e = \frac{1}{2}ke^2$. Use extension in m. Spring equations within proportional limit.

Elastic returns to original shape; inelastic leaves deformation. Measure force/extension with ruler and known loads.

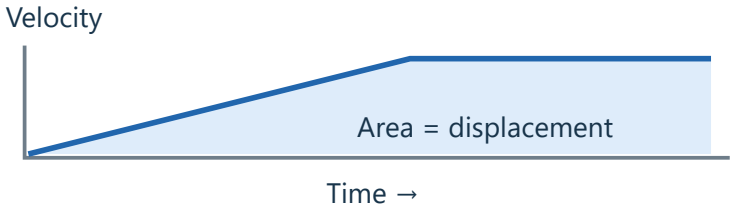
Turn & move

3 Moments & pressure

$M = Fd$, perpendicular d. Balance clockwise/anticlockwise moments. Levers/gears transmit turning.

$p = F/A$; $p = h\rho g$. Greater depth → greater p; pressure difference gives upthrust.

4 Motion & laws



$v = s/t$; $a = \Delta v/t$; $v^2 - u^2 = 2as$ for constant acceleration.

Distance–time slope = speed; velocity–time slope = a, area = displacement. **$F = ma$;** third-law forces act on different objects.

Stop & conserve

5 Stopping & falling

Stopping = thinking + braking. Thinking = vt; braking transfers kinetic energy to thermal stores.

Weight > drag → accelerates. Drag grows with speed until forces balance at terminal velocity.

6 Momentum

$p = mv$; $F = \Delta p/\Delta t$. Closed system: total momentum conserved. Use signed velocities.

Airbags/mats increase stopping time, reducing average force for the same momentum change.